



Figure 3: Components of Selenium (Image credits: antidisestablishmentarianism)

5. Capable of capturing test items or objects, both with or without the X Path values.
6. Synchronises the script with the mouse and keyboard related commands.
7. Selenium Grid dramatically speeds up functional testing of Web applications by leveraging the user's existing computing infrastructure.
8. Allows different users to easily run multiple tests in parallel, on multiple machines or even in a heterogeneous environment.
9. Can be deployed on Windows, Linux and Macintosh platforms.

Watir

Watir (Web Application Testing in Ruby, pronounced 'water') is a Ruby based automation tool for automating Web browsers. Watir was developed by Bert Petticoated and Paul Rogers. It actually drives an application on the Web in a manner that's similar to how users do. It clicks on a browser, fills the form data, presses buttons, etc. It also checks results such as when any text appears on a page. It is quite simple and flexible. Using the Ruby programming language, Watir can even make a connection to the database and read files (.text, .ls, .XML, etc). Users can also create their own functions and library for reusable code.

Supported scripting language: Ruby

Features:

1. Test scripts in Watir can be written using Ruby, and it consists of a Ruby library which helps to simulate users' action in the browser.
2. Is able to use the full power of Ruby to read the data from Excel files and write the data to Excel files or XML files in test scripts.
3. Very popular among testers; hence, the community that backs it is growing.
4. Supports all Web apps, no matter what language these are developed in.

5. Supports multiple browsers and that too, on different platforms.
6. Powerful and easy to use, yet wonderfully lightweight.

SoapUI

SoapUI is the world's most widely-used open source Web service testing application. It can test different Web services for service-oriented architectures (SOA) or representational state transfer (REST) APIs. It was developed by Liveware Software, which was later acquired by Smart Bear Software in 2011. SoapUI was initially released to Source Forge in September 2005. It is free software that is licensed under the terms of the European Union Public License (EU PL). It is developed entirely on the Java platform, and it uses Swing for the user interface. SoapUI is cross-platform and also supports Eclipse, IDEA and Net Beans. It can test SOAP and REST Web services along with JMS and AMF. SoapUI also provides extensive opportunities for scripting by using either Groovy or JavaScript (since Soap 3.0).

Supported scripting languages: Groovy Script, JavaScript
Features:

1. **Functional testing**
 - a. Allows all testers to write functional API tests in SoapUI.
 - b. Supports the drag-and-drop feature.
 - c. Also supports test debugging and allows data driven test development.
 - d. Allows advanced scripting (code can be customised depending on the scenario).
2. **Security testing**
 - a. Capable of performing a complete set of vulnerability scans.
 - b. Prevents SQL injection in order to secure the databases.
 - c. Scanning performed for cross site scripting.
 - d. Also performs boundary scans and fuzzing scans to avoid erratic behaviour of the services.
3. **Load testing**
 - a. Distributes the load tests across different load agents.
 - b. Simulates high volume and real-world load testing.
 - c. Advanced customised reporting to capture different performance parameters.
 - d. Allows end-to-end performance monitoring of the system.
4. **Supported protocols/technologies**
Supports most comprehensive protocols like SOAP, WSDL, AMF, HTTP, REST, JD and JMS.
5. **SOAPUI integration**
SoapUI can be easily integrated very well with popular tools like Maven, J Unit, Apache-Ant, etc.